EXPERIMENT 2: DATA DISCOVERY AND PREPARATION

LEARNING OBJECTIVES:

- To perform data discovery and exploratory data analysis (EDA) on a customer churn dataset.
- To analyze the distribution of categorical variables such as Churn, Gender, Contract Type, and Payment Method.
- To visualize numerical features such as MonthlyCharges, TotalCharges, and Tenure using histograms and boxplots.
- To identify and detect outliers using the Interquartile Range (IQR) method.
- To analyze the relationship between customer churn and important features such as Contract Type, Internet Service, Monthly Charges, and Tenure.
- To identify important patterns and factors that may influence customer churn.
- To prepare the cleaned dataset for machine learning by splitting it into 80% training and 20% testing data.

METHODS USED / PYTHON LIBRARIES INVOLVED:

Library/method	Purpose	Reason for Use
Pandas	Data loading, cleaning and manipulation	Efficiently handles tabular datasets
NumPy	Numerical operations	Useful for numerical calculations
Matplotlib	Data visualization	Creates histograms, boxplots and charts
Seaborn	Statistical visualization	Creates attractive count plots and relationship plots
Scikit-learn	Train-test splitting and preprocessing	Provides machine-learning preprocessing functions

PROGRAM CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split

df = pd.read_csv("Customer_Churn.csv")
df["TotalCharges"] = df["Subscription_Length"] * df["Monthly_Spend"]

print(df.head())
print(df.shape)
print(df.isnull().sum())
print(df.describe())

for c in ["Subscription_Length","Monthly_Spend","TotalCharges"]:
    sns.histplot(df[c], kde=True)
    plt.title(c)
    plt.show()

    sns.boxplot(y=df[c])
    plt.show()

    q1,q3=df[c].quantile([.25,.75])
    iqr=q3-q1
    print(c,"Outliers:",len(df[(df[c]<q1-1.5*iqr)|(df[c]>q3+1.5*iqr)]))

sns.countplot(x="Churn",data=df)
plt.show()

sns.boxplot(x="Churn",y="Monthly_Spend",data=df)
plt.show()

sns.boxplot(x="Churn",y="Subscription_Length",data=df)
plt.show()

print(df.groupby("Churn")[["Subscription_Length","Monthly_Spend","TotalCha
rges"]].mean())

X=df.drop(columns=["Customer_ID","Churn"])
```

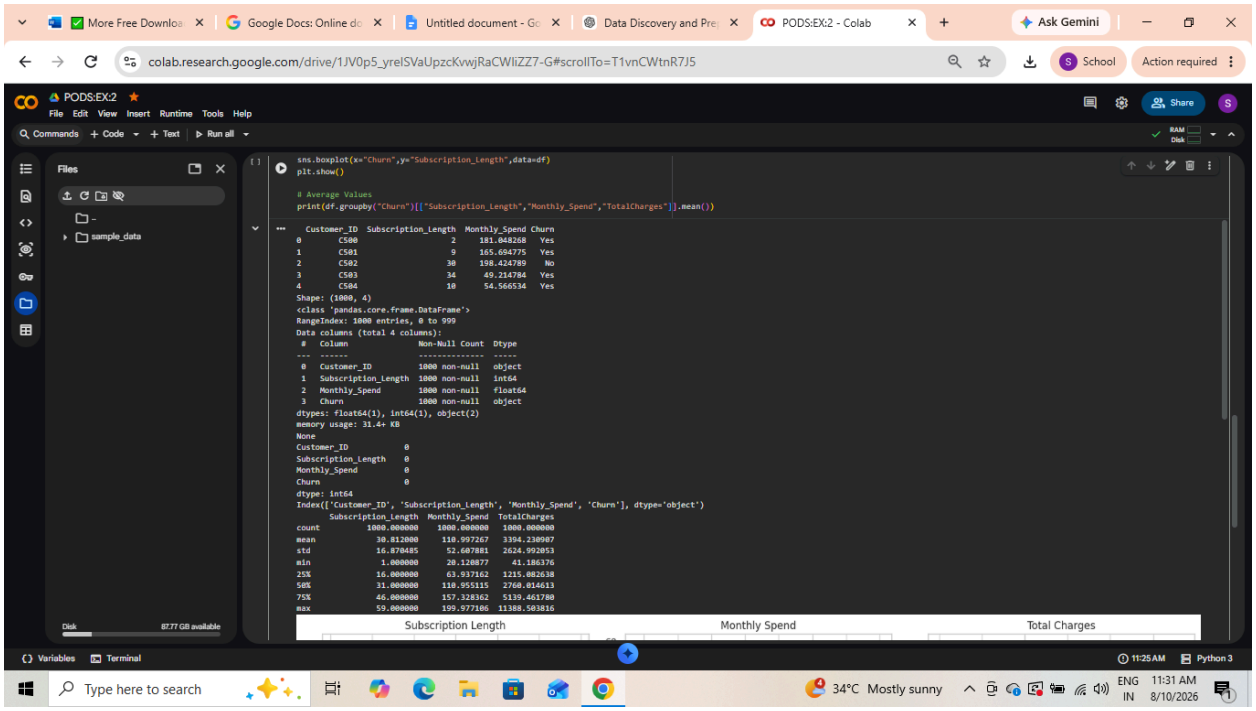
```
y=df["Churn"]

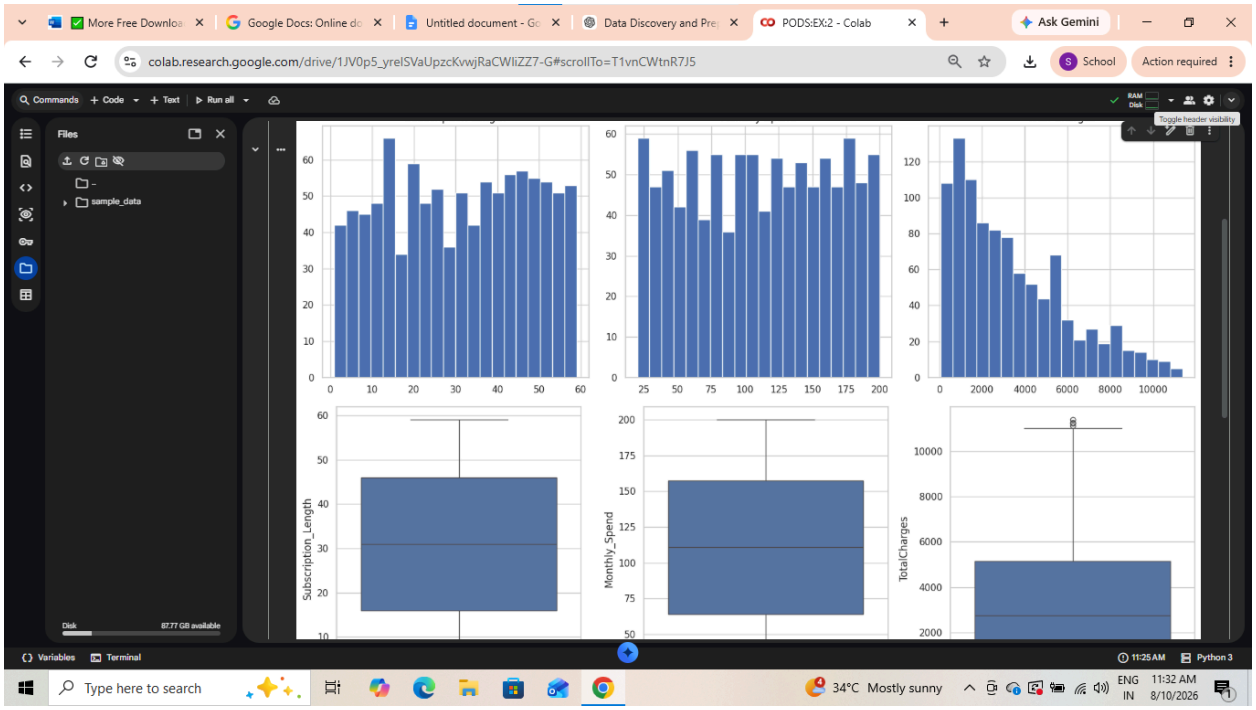
X_train,X_test,y_train,y_test=train_test_split(
    X,y,test_size=.2,random_state=42
)

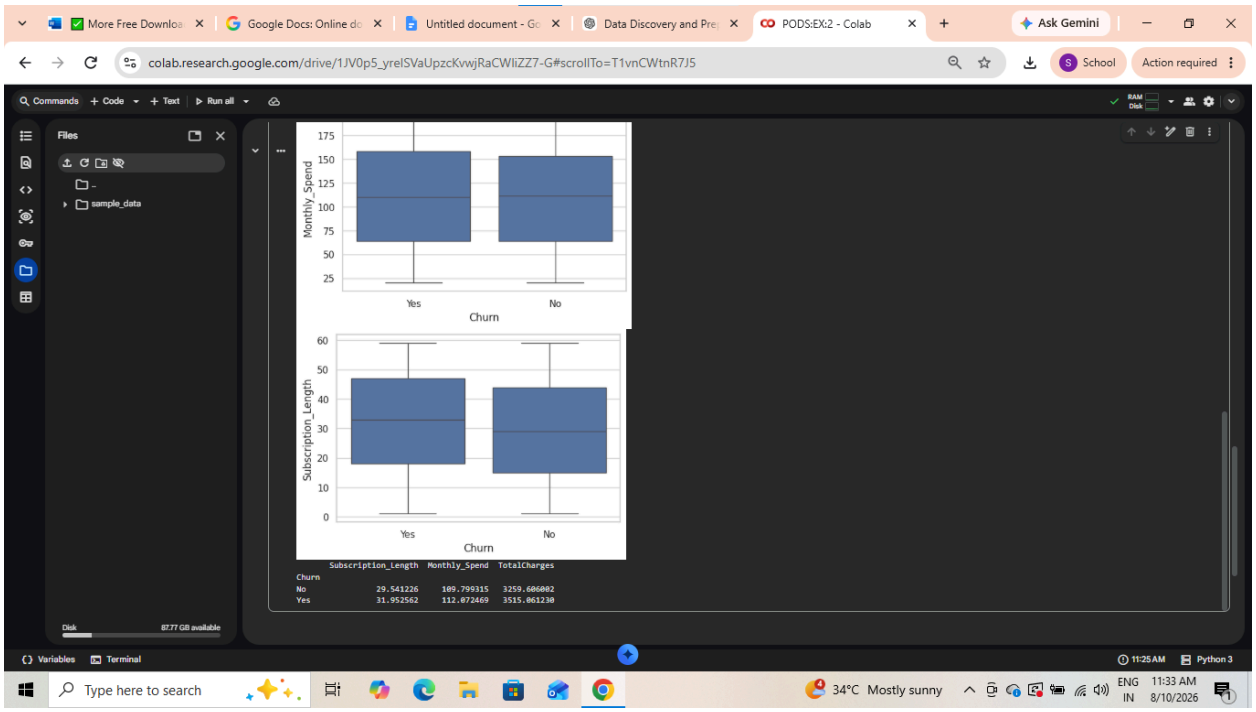
df.to_csv("CustomerChurn_Preprocessed.csv",index=False)

print("Train:",X_train.shape)
print("Test:",X_test.shape)
print("Saved successfully")
```

OUTPUT:




PODS:EX:2 - Colab
colab.research.google.com/drive/1JV0p5_yreISVaUpzcKvwjRaCWIiZZ7-G#scrollTo=T1vnCWtnR7J5
Subscription_Length
Monthly_Spend
TotalCharges
34°C Mostly sunny
11:32 AM
8/10/2026


PODS:EX:2 - Colab
colab.research.google.com/drive/1JV0p5_yreISVaUpzcKvwjRaCWIiZZ7-G#scrollTo=T1vnCWtnR7J5
Monthly_Spend
Churn
Yes
No
Subscription_Length
Churn
Yes
No
Subscription_Length Monthly_Spend TotalCharges
Churn
No 29.541226 109.799315 3259.606002
Yes 31.952562 112.072469 3515.861230
34°C Mostly sunny
11:33 AM
8/10/2026

LEARNING OUTCOMES:

- Understand data discovery and exploratory data analysis.
- Analyze customer churn and behavior.
- Create histograms, boxplots, and count plots.
- Detect outliers using the IQR method.
- Identify features related to customer churn.
- Split data into training and testing sets.
- Prepare data for predictive analytics.